

Discrete Mathematics

Computing and Mathematics

May, 2026

Discrete Mathematics (A10841)

Short Title:	Discrete Mathematics	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Mathematics and Physics	
Author	KMURPHY	
Credits	5	Level: Introductory

Description of Module / Aims

This module provides a solid foundation of selected topics in discrete mathematics related to computing and information sciences. The topics are covered in an elementary manner in order to reinforce understanding of concepts and improving algebraic problem-solving skills so that the student can effectively proceed with their study of a degree programme in computing.

Programmes

MATH-0001	BSc (Hons) in Applied Computing (International) (WD_KACMI_B)	2 / 3 / M
MATH-0001	BSc (Hons) in Applied Computing (WD_KACCM_B)	1 / 1 / M
MATH-0001	BSc (Hons) in Applied Computing (WD_KCOMP_B)	1 / 1 / M
MATH-0001	BSc (Hons) in Computer Forensics and Security (WD_KCOFO_B)	1 / 1 / M
MATH-0001	BSc (Hons) in Computer Science (WD_KCMSC_B)	1 / 1 / M
MATH-0001	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	2 / 3 / M

Indicative Content

- Sets, relations and functions: Venn diagrams; set operations and their laws; properties of relations; function domain, range and target set; function composition; function definition using lookup table, direct formula, and recursion
- Proofs and predicate logic: Propositions; logical connectives and truth tables; informal arguments and formalising arguments; introduction to predicates and quantifiers
- Proof techniques: direct proof; induction; proof by contradiction and counterexamples
- Enumeration: combinatorial enumeration problems; permutations and combinations; binomial expansions
- Graph Theory: Basic definitions; graph traversal techniques and applications
- Recurrence Relations: sigma and product notation; finite sequences and recurrence relations; iteration techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate competency in algebraic manipulation of expressions involving sets, combinatorials, predicate logic elements and finite sequences.
2. Use mathematical reasoning to comprehend and construct mathematical arguments (direct proof, proof by contradiction, mathematical induction).
3. Employ elementary combinatorial techniques to resolve simple problems.
4. Use basic concepts in graph theory to represent and solve problems arising in computer science.
5. Compute basic recurrence relations and construct a recurrence relation when given a descriptive problem.
6. Apply concepts and methods in discrete mathematics to problems using a computer.

Learning and Teaching Methods

- Delivery of the subject will be through a mixture of lecture, tutorials and computer based practical sessions.
- The lectures will be used to introduce the context for the material, develop theory and lead students through worked examples.
- The tutorials will provide a forum through which the student will rehearse/refine the demonstrated skills. Students will be encouraged, through the problem sheets, to construct valid and precise mathematical arguments and will be expected to produce solutions using appropriate mathematical notation.
- The practical programme is designed to re-enforce the strong interconnections between discrete mathematics and Computer Science concepts through the use of CAS and guided development of students own code. A typical activity consists of investigating the effect on computation time of different recursive and iterative implementation of a recurrence relation solver and their role in computer science.
- Active engagement with frequent practise on examples is strongly encouraged through regular course work and formative class tests.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	25%	
<i>Practical</i>	25%	6
Final Written Examination	75%	1,2,3,4,5

Assessment Criteria

<40%: Inability to demonstrate knowledge or understanding of the main concepts in discrete mathematics as outlined in the syllabus content, inability to apply concepts to selected problems.

40%-49%: Able to demonstrate a basic understanding of the fundamental concepts and techniques in vector calculus and matrix systems as concepts outlined in syllabus content.

50%-59%: In addition to above, using appropriate mathematical notation.

60%-69%: All the above, in addition be able to determine appropriate mathematical techniques to analyse applied problems and to express their work with rigour and precision.

70%-100%: All the above to an excellent level. Demonstrates an ability to put a solution into a context and assess whether such solutions are meaningful.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Practical	12	
Independent Learning	75	

Supplementary Material

- Johnsonbaugh, R. _Discrete Mathematics_. 7th. NY: Macmillan, 2008.

Requested Resources

- Room Type: Computer Lab test 14